Computer Science & IT

Theory of Computation



CFL

Lecture No. 3



By- Ravindra Sir

Recap of Previous Lecture



- Topic DPADA
- Topic
- Topic

Topics to be Covered



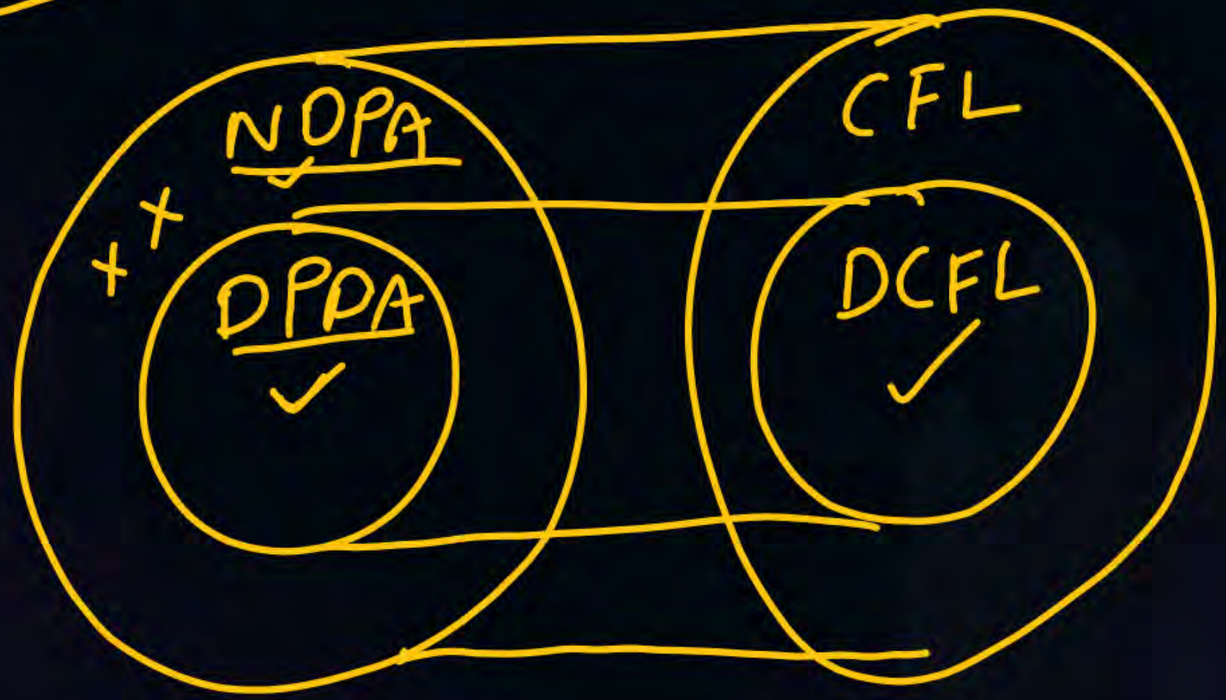
- Topic
- Topic
- Topic

NDPDA

NFA = DFA

DPDA < NDPA

DTM = NTM



Changing Batteries please wait

Non deterministic PDA

$WW^R / w \in (a+b)^+ \rightarrow$

abba

aabbaa

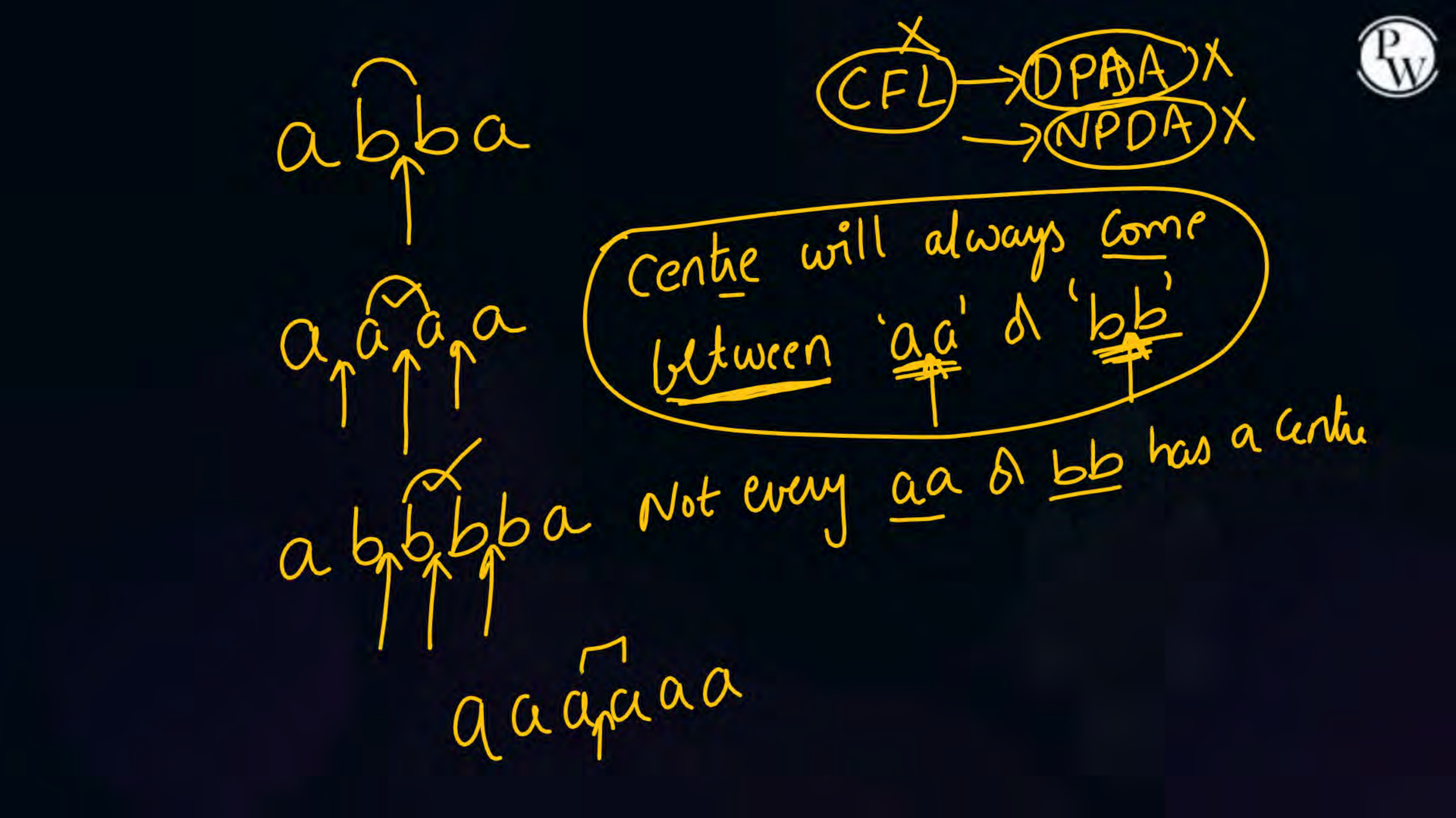
DPDA

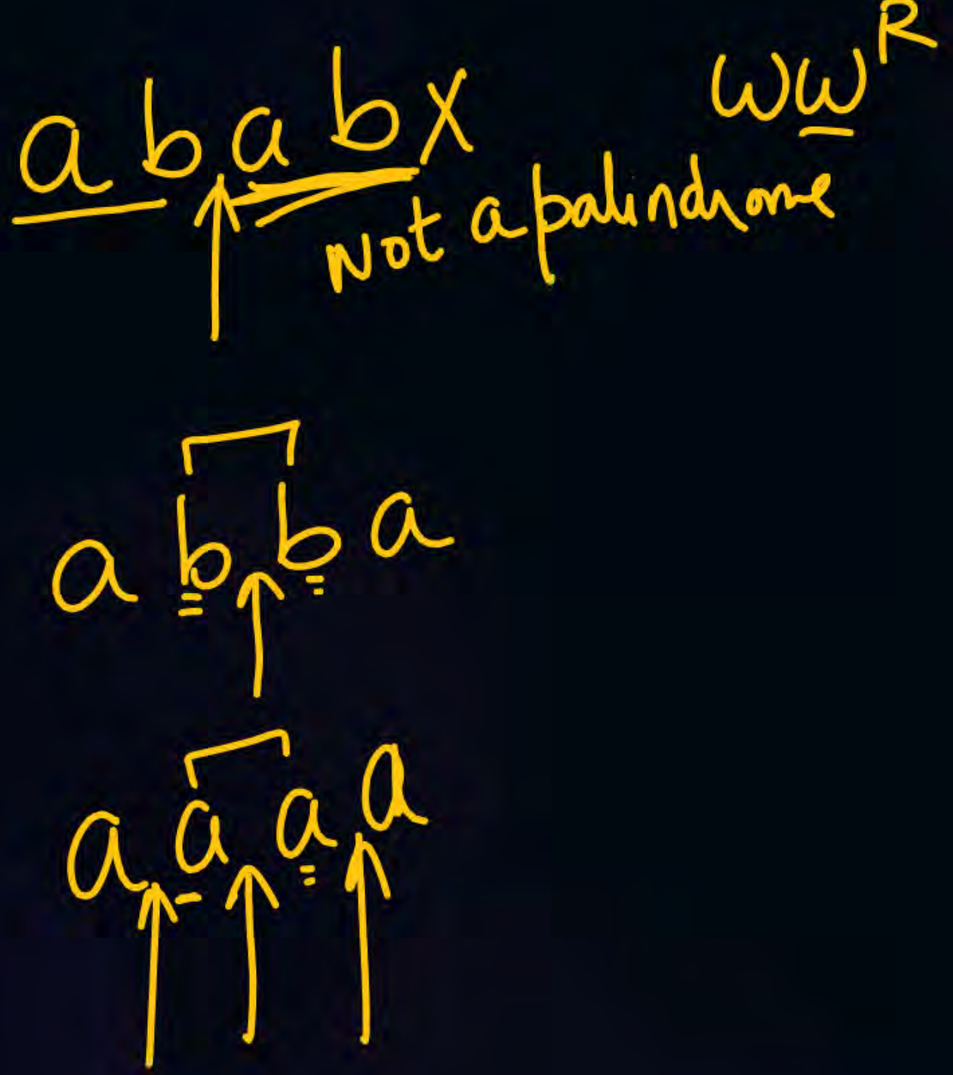


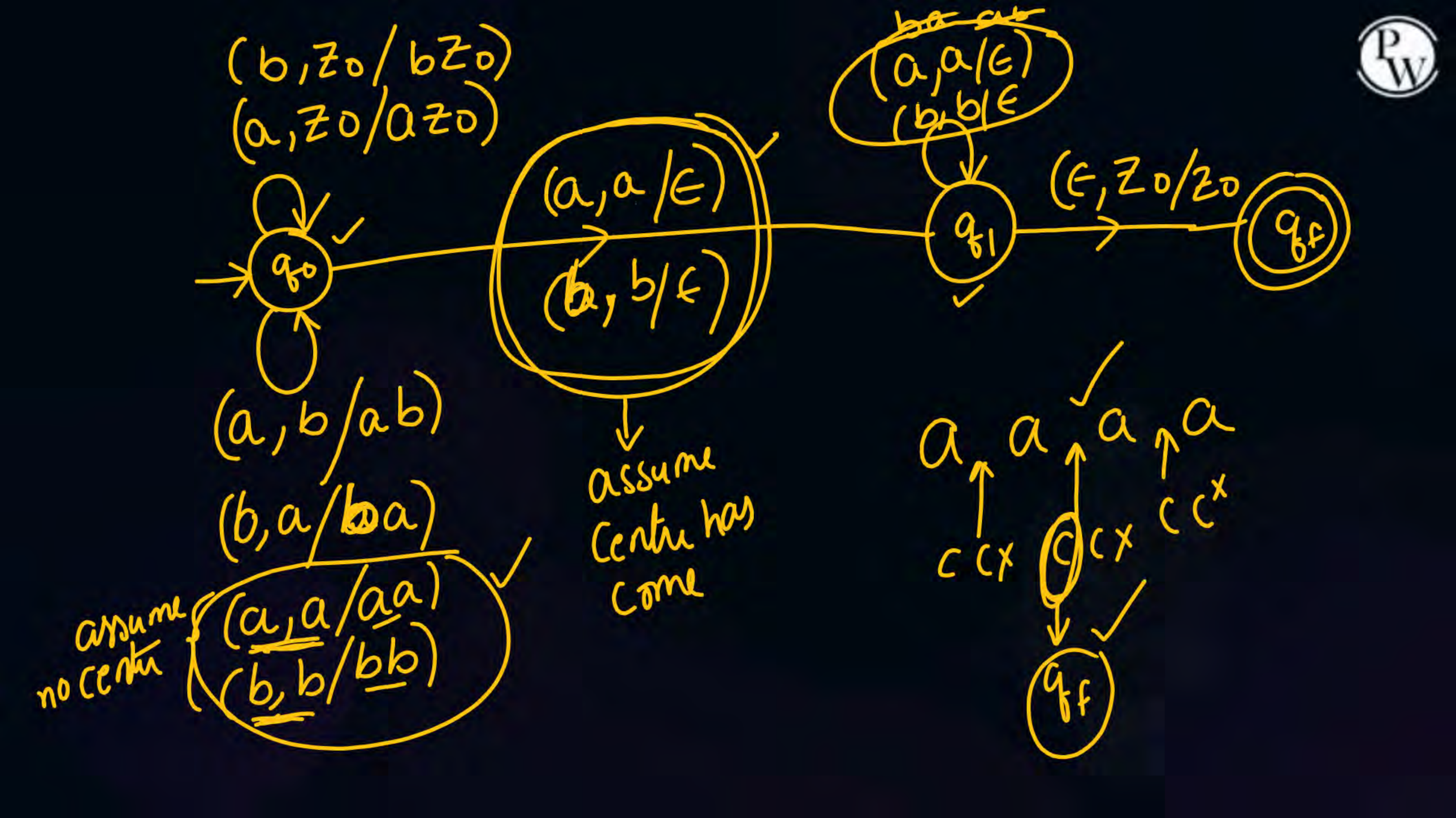
We cannot give a DPDA because we dont know centre

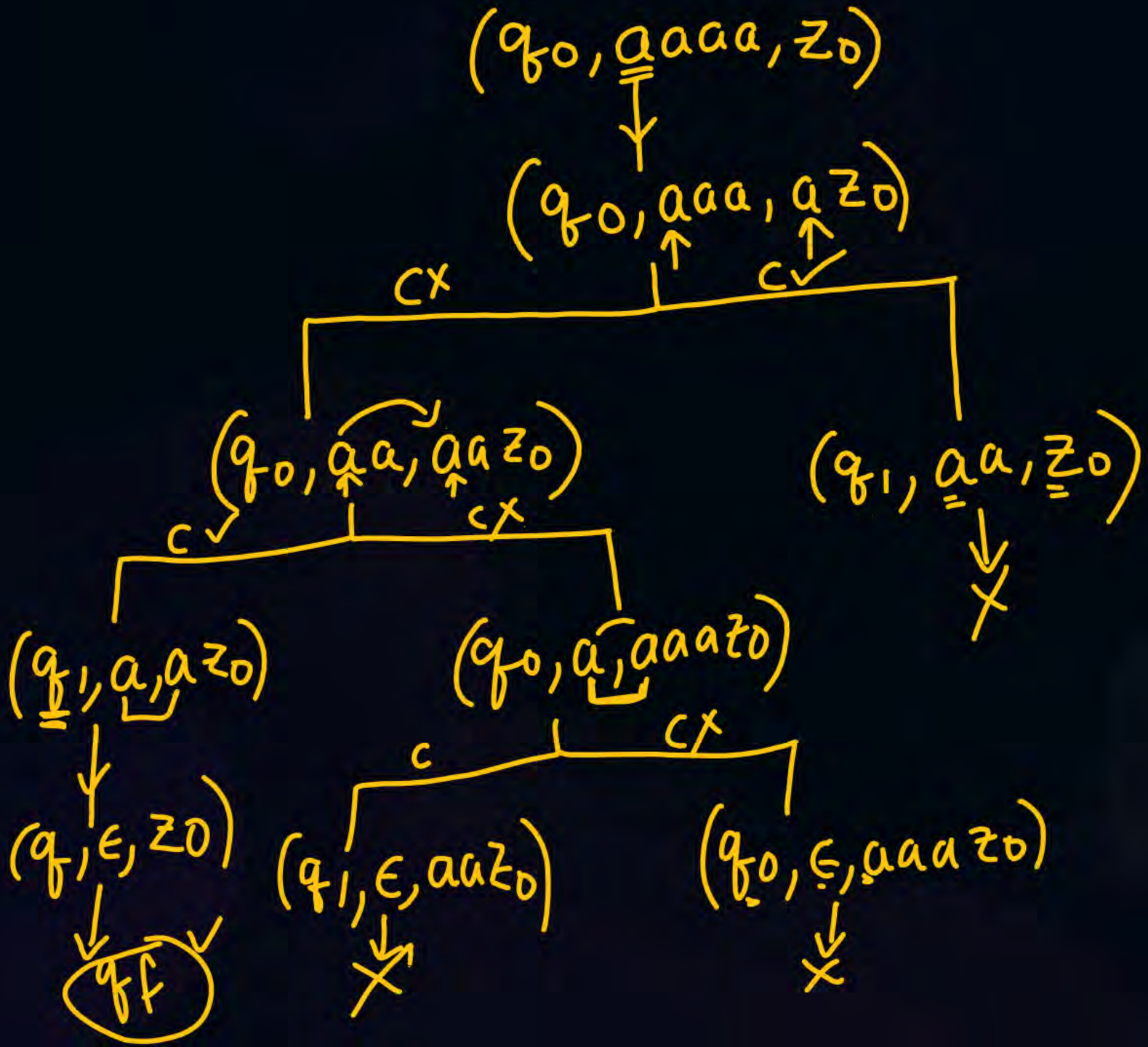
WCW^R → upto C, push

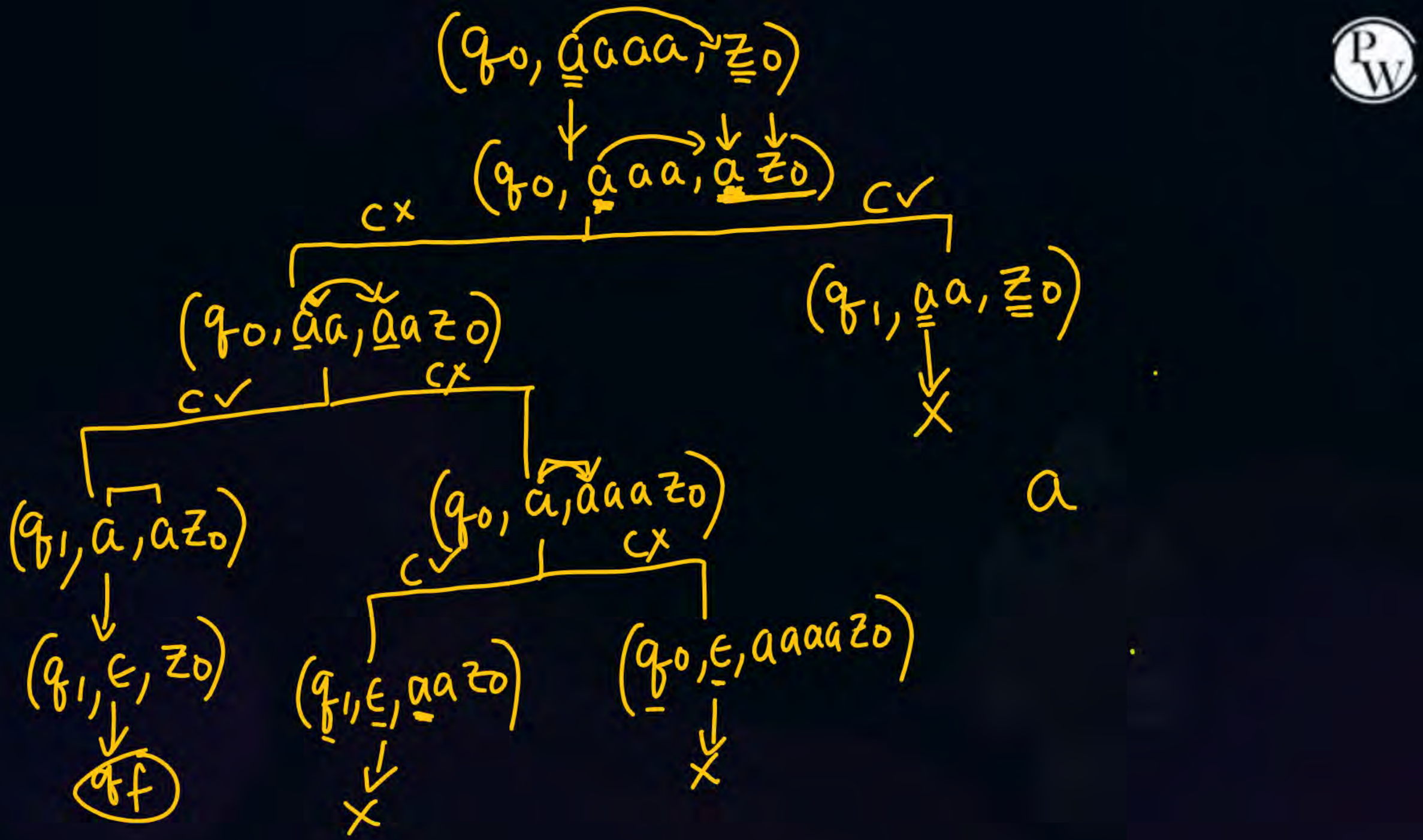
after C, → pop

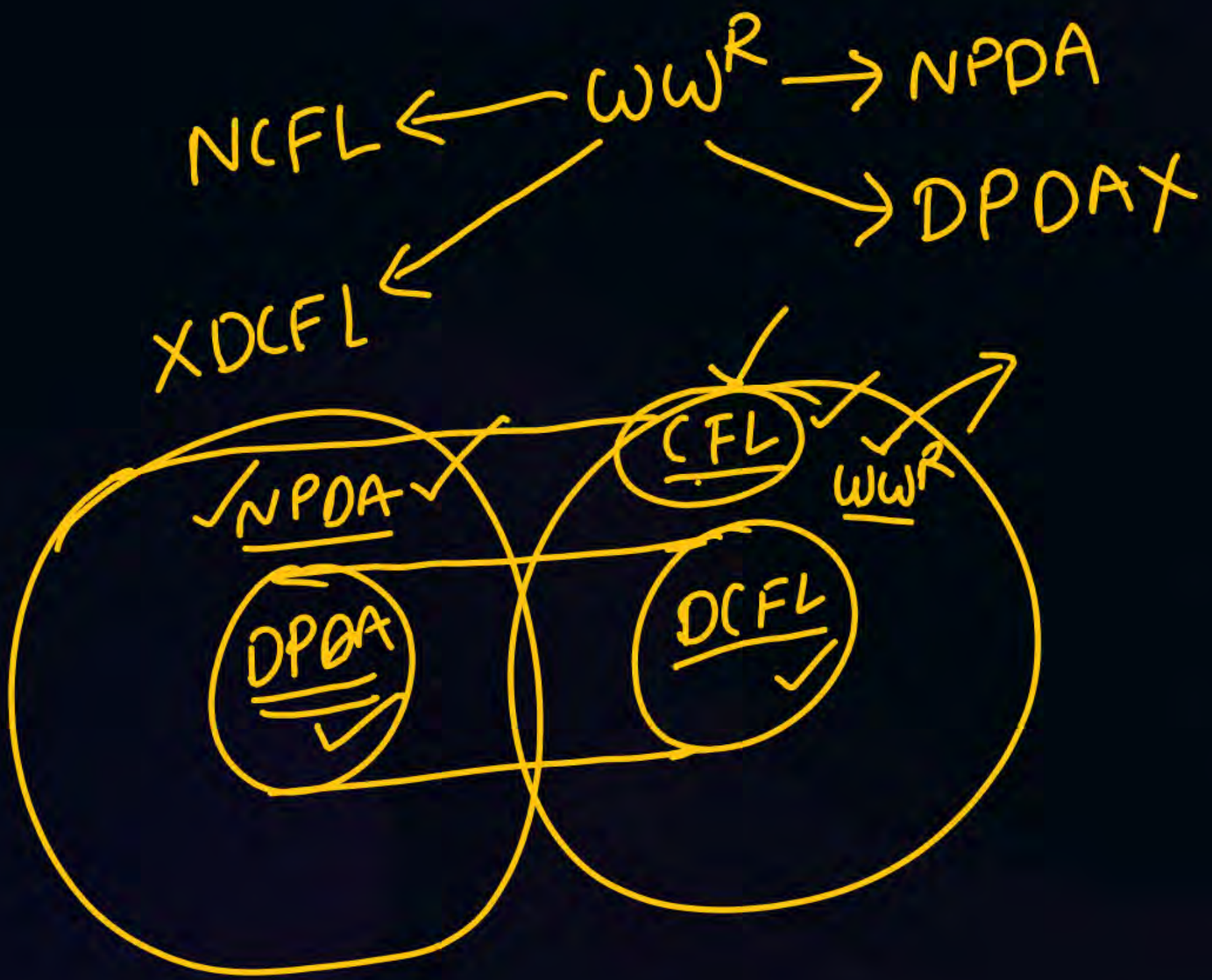

CFL
DPDA
NPDA
abba
aaaa
abbbba
aaaaaaa
Centre will always come between 'aa' & 'bb'
Not every aa & bb has a centre


abaabX
ww^R
Not a palindrome
abba
aaaa

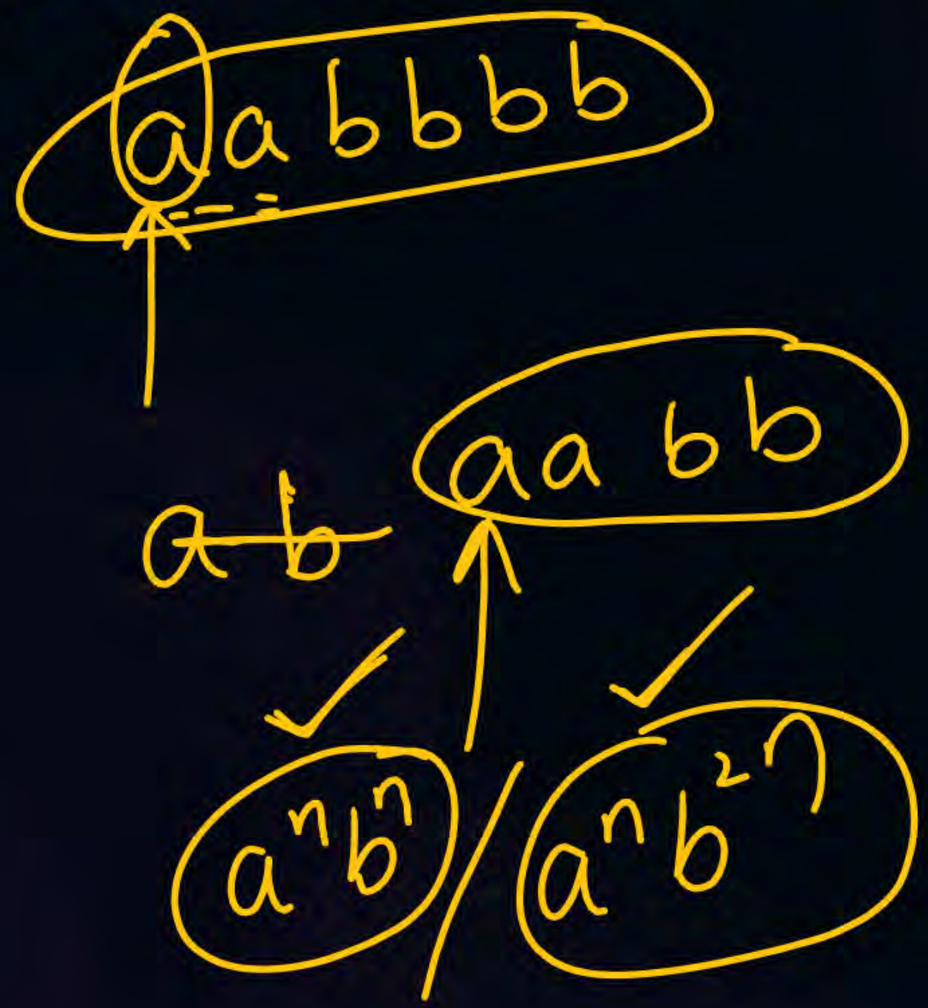

(b, Z0 / bZ0)
(a, Z0 / aZ0)
(a, a / ∈)
(b, b / ∈)
(a, a / ∈)
(b, b / ∈)
(∈, Z0 / Z0)
q0
q1
qf
(a, b / ab)
(b, a / baa)
(a, a / aa)
(b, b / bb)
assume no centre
assume centre has come
a a a a a
c cx cx cx
qf

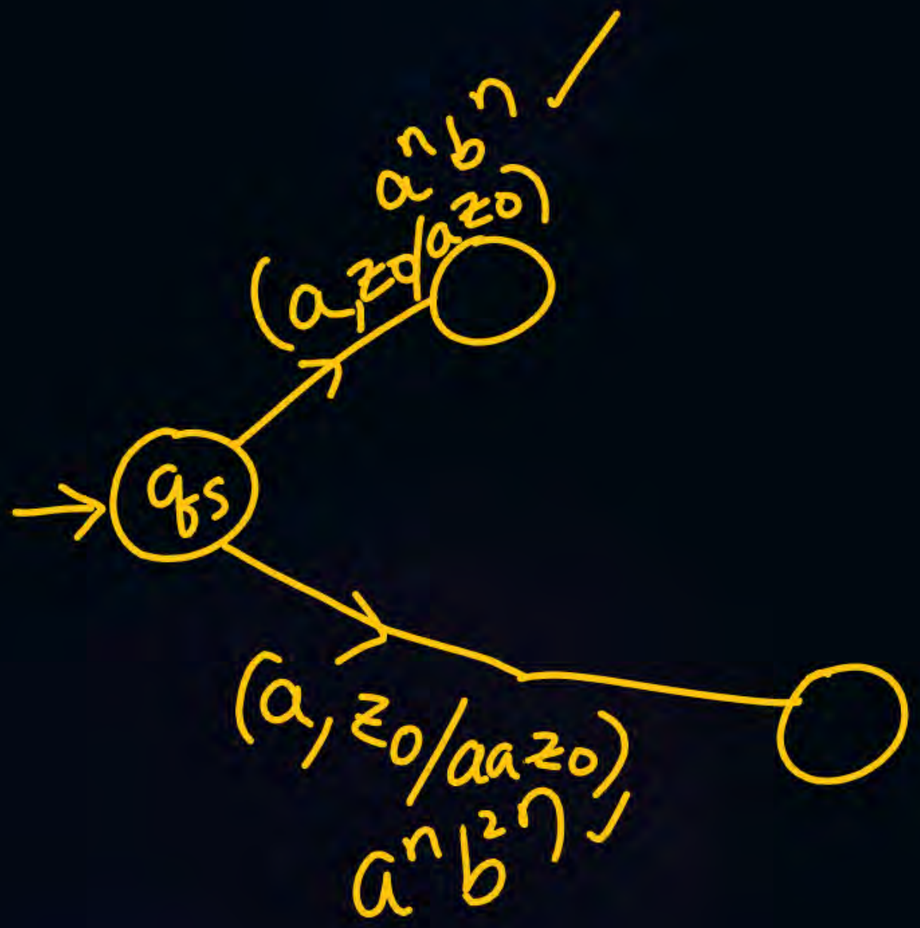

(q0, aaaa, Z0)
(q0, aaa, aZ0)
CX
C✓
(q0, aa, aaZ0)
(q1, aa, Z0)
C✓
CX
(q1, a, aZ0)
(q0, a, aaaZ0)
(q1, ε, Z0)
qf
C
CX
(q1, ε, aaZ0)
(q0, ε, aaaZ0)

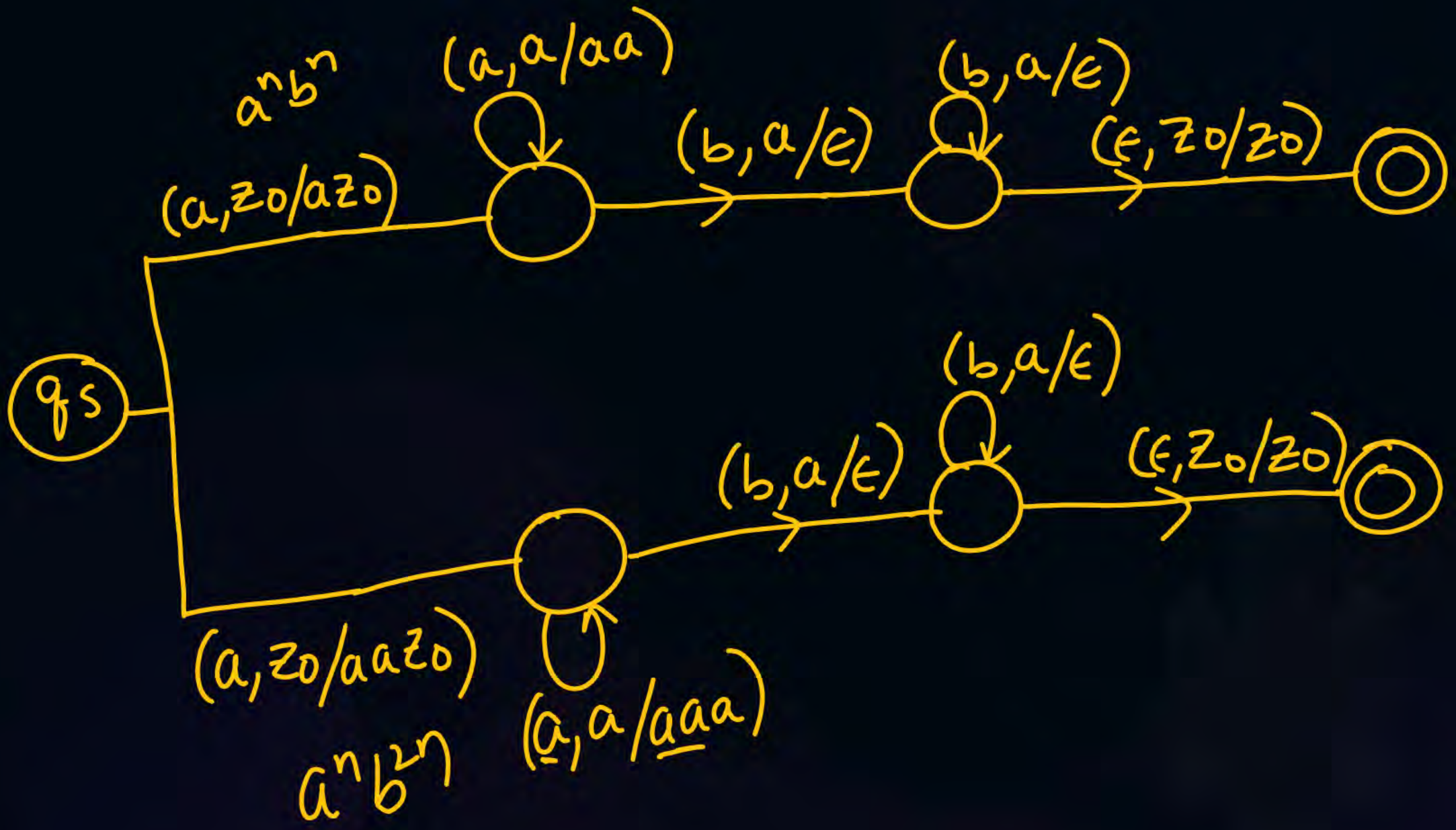

(q0, aaaa, z0)
(q0, aaaa, az0)
c✗
c✓
(q1, aa, z0)
X
(q0, aa, aaz0)
c✓
c✗
(q1, a, az0)
(q0, a, aaaz0)
a
(q1, ε, z0)
c✓
c✗
qf
(q1, ε, aaz0)
(q0, ε, aaaaz0)
X
X


NCFL
WW^R
NPDA
DPDAX
XDCFL
NPDA
CFL
WW^R
DPDA
DCFL

$L = \{a^n b^n / n \geq 1\} \cup \{a^n b^{2n} / n \geq 1\}$




$a^n b^n$ ✓
$(a, z_0 / a z_0)$
q_s
$(a, z_0 / aa z_0)$
$a^n b^{2n}$


$a^n b^n$
(a, a/aa)
(b, a/ϵ)
(b, a/ϵ)
(ϵ, Z0/Z0)
(a, Z0/aZ0)
qs
(b, a/ϵ)
(b, a/ϵ)
(ϵ, Z0/Z0)
(a, Z0/aaZ0)
$a^n b^{2n}$
(a, a/aaa)

Limitation of DPDA :-

1) when to push and when to pop is not clear, the DPDA fails

Ex :- Even length palindrome
Odd length palindrome
all palindrome

$$WcW^R \longrightarrow \text{DPDA}$$



2) How much to push is not clear, then it is not possible by DPDA

NDPA { $a^n b^n \cup a^n b^{2n} / n \geq 1$

$w / n_a(w) = n_b(w)$ (✓) & $n_a(w) = 2 * n_b(w)$ (✓)

In the above questions we dont know how may a's to push

$a^n b^n \cup a^n b^{2n} \cup a^n b^{3n} \cup a^n b^{4n}$

↓ NPDA

3) more than one comparision → DPDA fails

$a^m b^n c^p / m=n \text{ or } m=p$ ✗DCFL ✓CFL ✓NCFL



$a^m b^n c^p / m=n$ and $m=p$ ✓CSL ✗CFL

$w / n_a(w)=n_b(w)$ & $n_a(w)=n_c(w)$ ✓CFL ✗DCFL



$***$

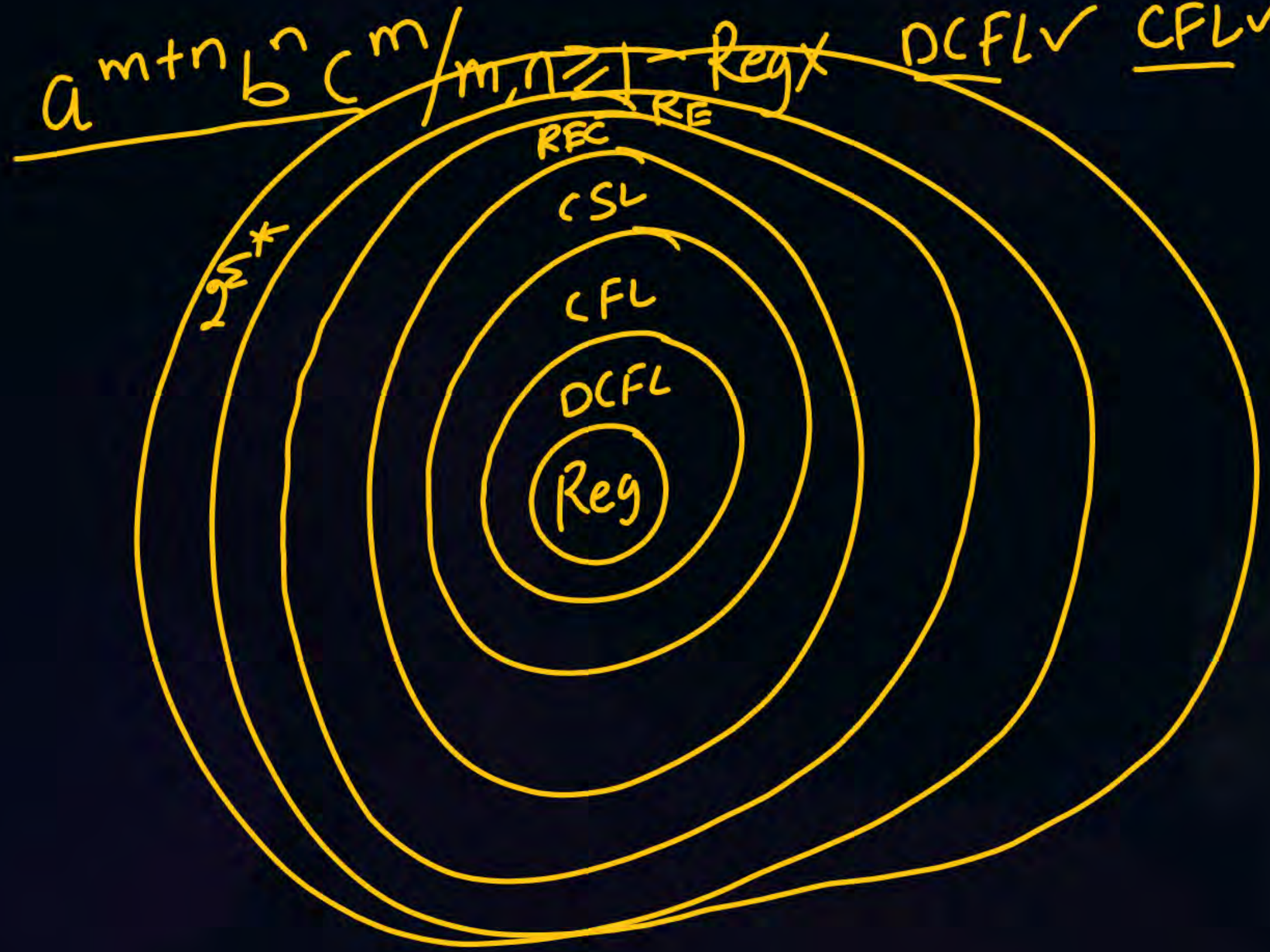
$a^m b^n / n \leq m \leq 2n$

✓CFL

XDCFL

Reg, DCFL, CFL, CSL

$a^{m+n}b^n c^m / m,n \geq 1$ — Reg ✗ DCFL ✓ CFL ✓ CSL ✓



$a^m b^{m+n} c^n / m,n \geq 1$ Regx DCFL✓ CFL✓ CSL✓ RecL✓ RE✓

$\frac{a^m}{Pu}\frac{b^m}{Po}\frac{b^n}{Pu}\frac{c^n}{Po}$

Strongest

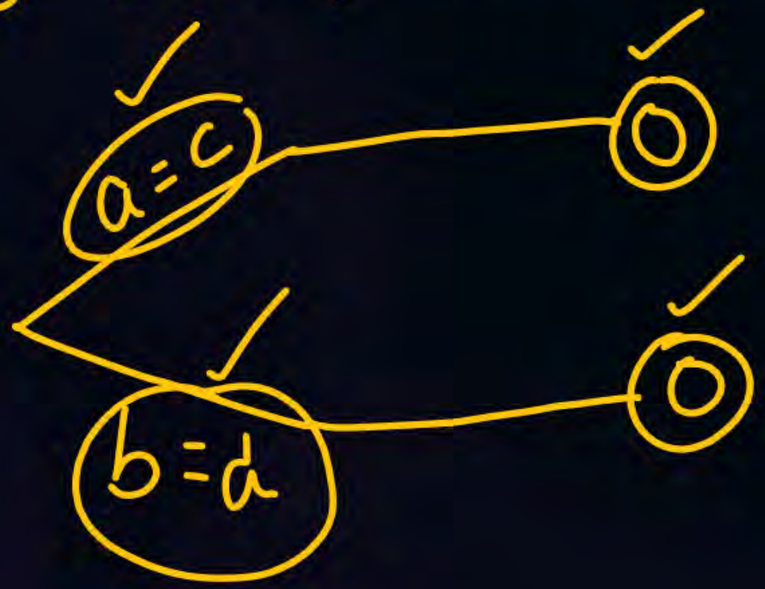
$a^m b^n c^{m+n} / m,n \geq 1 \rightarrow$ Regx DCFL✓ CFL✓ CSL✓

$\frac{a^m}{P}\frac{b^n}{P}\frac{c^n}{Po}\frac{c^m}{Po}$

$a^m b^m c^n d^n / n, m \geq 1$ Regx DCFL✓ CFL✓ CSL✓

$a^m b^n c^m d^n$ Regx DCFLX CFLX CSL✓

$a^n b^m c^l d^k / n = l$ and $m = k$



$a^m b^n c^n d^m / n, m \geq 1$ Reg ✗ DCFL ✓ CFL ✓ CSL ✓

$a^m a^i c^m d^k / m, i, k \geq 1$

$a^{m+i} c^m d^k / m, i, k \geq 1$

$a^x c^y d^k / x > y, k \geq 1$

DCFL

$a^m b^n / m > n$ ✗ Reg DCFL ✓ CFL ✓ CSL ✓

$a^m b^n / m < n \rightarrow$ DCFL ✓

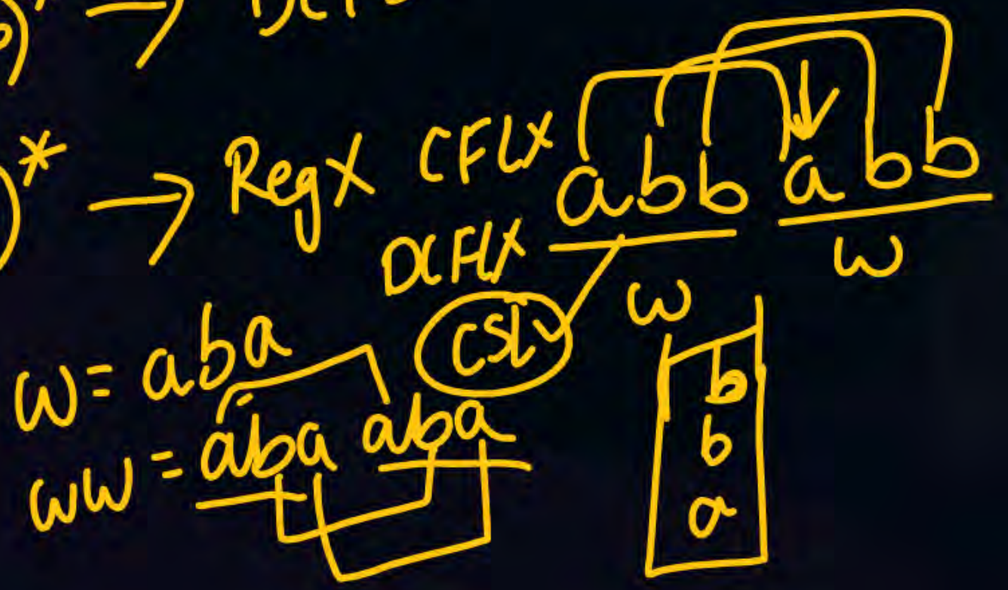
$a^n b^{2n} / n \geq 1 \rightarrow$ DCFL ✓

$a^n b^{n^2} / n \geq 1 \rightarrow$ CSL

$ww^R / w \in (a+b)^* \rightarrow$ DCFL ✗ CFL ✓

$wcw^R / w \in (a+b)^* \rightarrow$ DCFL ✓

$ww / w \in (a+b)^* \rightarrow$ Reg ✗ CFL ✗ DCFL ✗ CSL ✓



$a^n b^n c^m / n>m$

a^n b^n c^m

Push Pop

CSL ✓

CFL ✗

DCFL ✗

Reg ✗

$a^n b^i c^m / n=i$ and $i>m$

aabcc

aaϕϕϕ

δ

OR

AND

$a^n b^n c^m / n>m$

2 a's

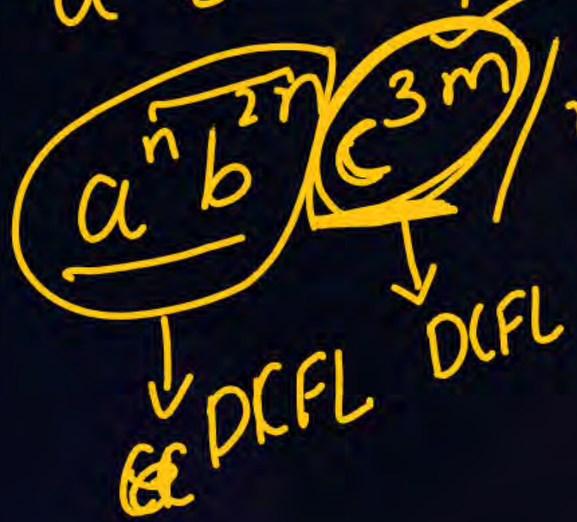
aabcc

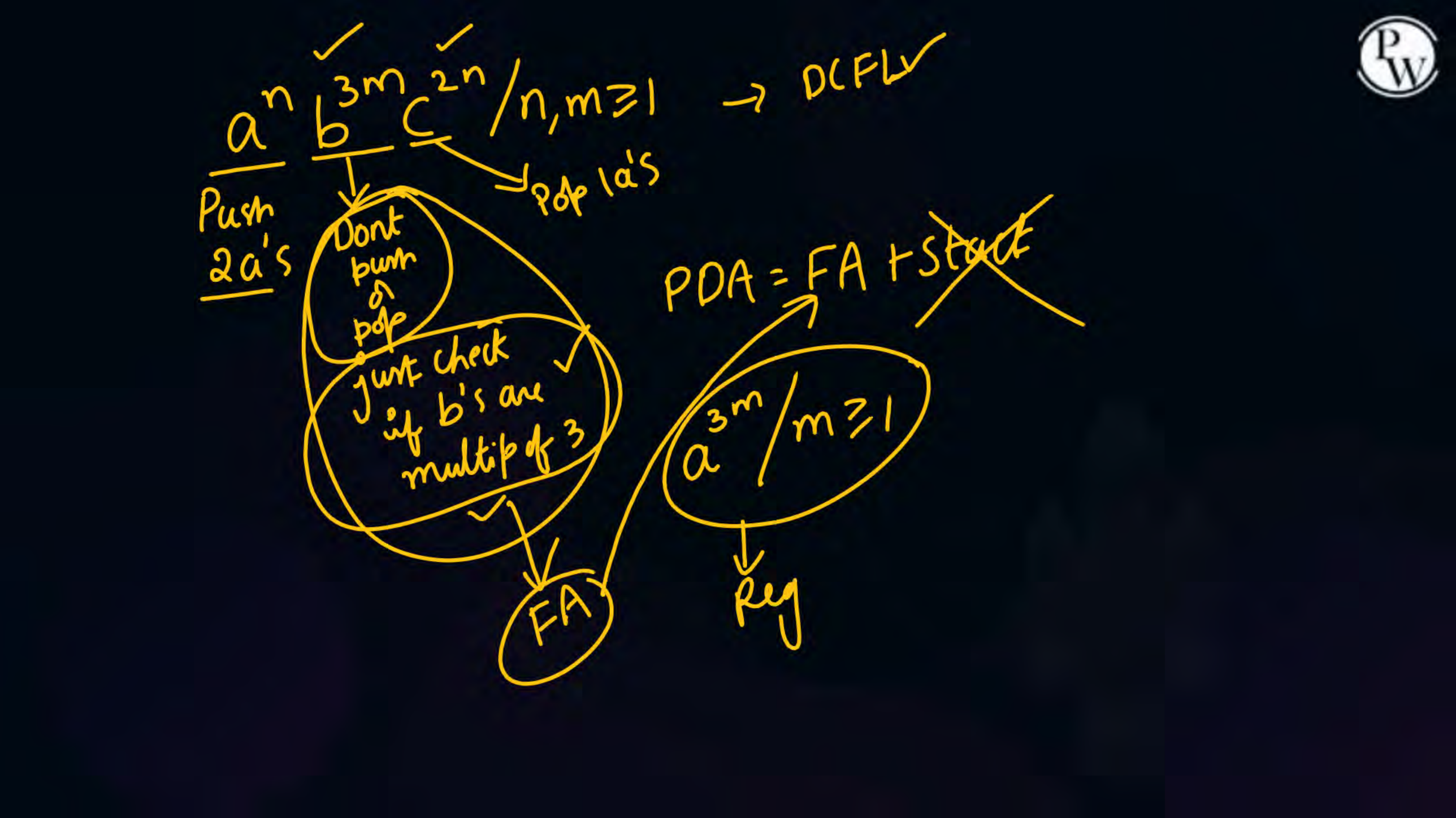
aaϕϕϕ

$a^n b^n c^n d^n / n \leq 10^{10}$

finite → Reg

$a^n b^{2n} c^{3n} / n \geq 1$ Reg✗ DCFL✗ CFL✗ CSL✓




$a^n b^{3m} c^{2n} / n, m \geq 1$ → DCFL
Push 2a's
Pop 1a's
Dont bum or pop
just check if b's are multiple of 3
FA
PDA = FA + Stack
$a^{3m} / m \geq 1$
Reg

$xcy \quad x, y \in (0+1)^*$

$(0+1)^* c (0+1)^* \rightarrow$ Reg DLFL, CFL, CSL

$xcx / x \in (0,1)^*$

$ww \rightarrow$ CSL

$wcw \rightarrow$ CSL

XCFL

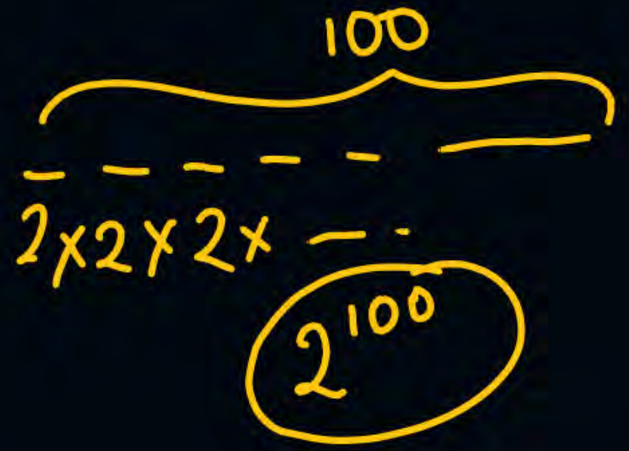
$(0+1)^* c (0+1)^* x$



$WW^R / w \in (a,b)^*$ $|w| = 100$ → finite → Reg

$WWW^R / w \in (a,b)^*$ $|w| = 100$ →

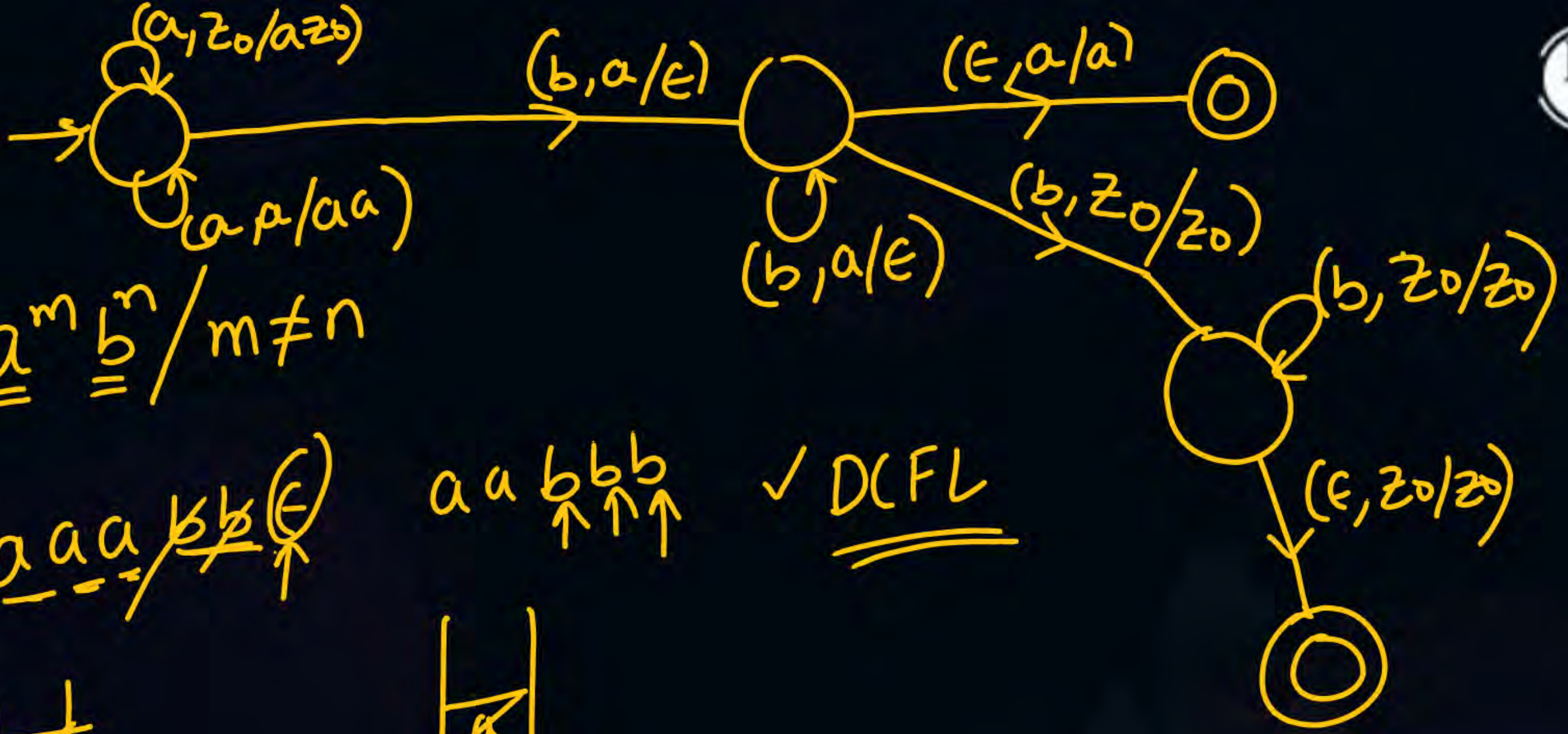
$WWW^R / w \in (a,b)^*$



↘ CSL ✓

XCFL

XDCFL



$a^m b^n / m \neq n$

a a a b b ε

a a b b b

✓ DCFL





$a^i b^j / i \neq 2j+1$ → DCFL ✓

$a^m b^n / m = 2n+1$ → DCFL

THANK - YOU